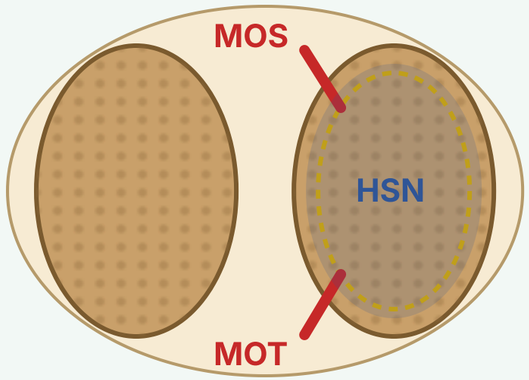
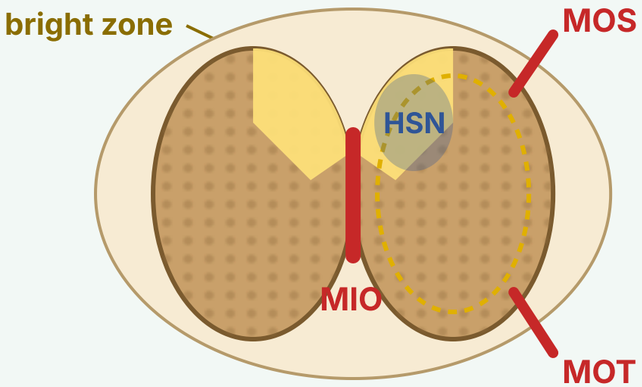
a The natural experiment

one genome, two visual machines



♀ hover, feed, lay eggs

2 retinal muscles · broad HSN field



♂ pursue mates

3 retinal muscles incl. the male-only MIO

bright zone · narrow HSN field

models on robots

b Modelling ladder

H1

WP1 · connectome → spiking circuit

Connectomes: ♀ FlyWire + ♂ MaleCNS
chemical synapses, cell types, morphology

MESH: electrical synapses
membrane appositions → EM proofreading

fit by differentiable simulation

Tier 1 · biophysical
gap-junction conductance ∝ contact area

reduce: KAN → equations

Tier 2 · reduced
LIF + gap junctions; ≥10× fewer variables

train: surrogate gradients

Tier 3 · trainable SNN
connectome-masked, task-trainable

motor readout
muscle pull → retinal shift; efference copy

Controversial-stimulus engine
differentiable detectors; stimuli on which competing models disagree most → c, e

predict

data

c Test in the animal

H2 · H2b

WP2 · Eristalis, MPI-BI + LMU

- ♂ vs ♀, both eyes: MIO as vergence muscle?
- tethered ♂/♀: retinal tracking, EMG of MOT, MOS and MIO; neurobiotin fills
- high risk: retinal tracking in free flight
- origin of the MIO: development and genes; the motor neuron as inducer (LMU)

d Explain

H3

WP3 · why dimorphic?

- same wiring prior and energy budget, trained for pursuit (♂ task) or stabilisation (♀ task)
- do bright-zone integration and a third effector emerge only for pursuit?
- ablation, mean-field, KAN readouts → motifs

e Transfer to machines

H4

WP4 · active event-based vision

- two event cameras on micro-movement stages; fixed vs fly-derived vs learned policies
- motif SNNs on neuromorphic chips (sub-mW)
- benchmarks vs matched deep networks: small targets · ego-motion · depth · quadrotor

f Validate at scale, with people

WP5 · courses, robot zoo, community proofreading, hoverfly watch

IU computer-vision courses
students build one motif per semester; remote students get kits by mail

Robot zoo
♂-, ♀- and fly-type circuits on low-cost camera robots: thousands of real-world trials

Community proofreading
citizens check candidate electrical synapses in the MESH viewer → b

Hoverfly watch
smartphone videos of ♂ pursuits in gardens → task statistics for d